

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

What's inside

- Global traffic growth and major connectivity shifts
- Top Internet services and the rise of generative AI
- Protocol + crypto adoption (HTTP/3, PQ, TLS 1.3)
- Disruptions + security threats shaping 2025 outlook

17% Global Traffic Growth and Starlink's 3.3x Surge

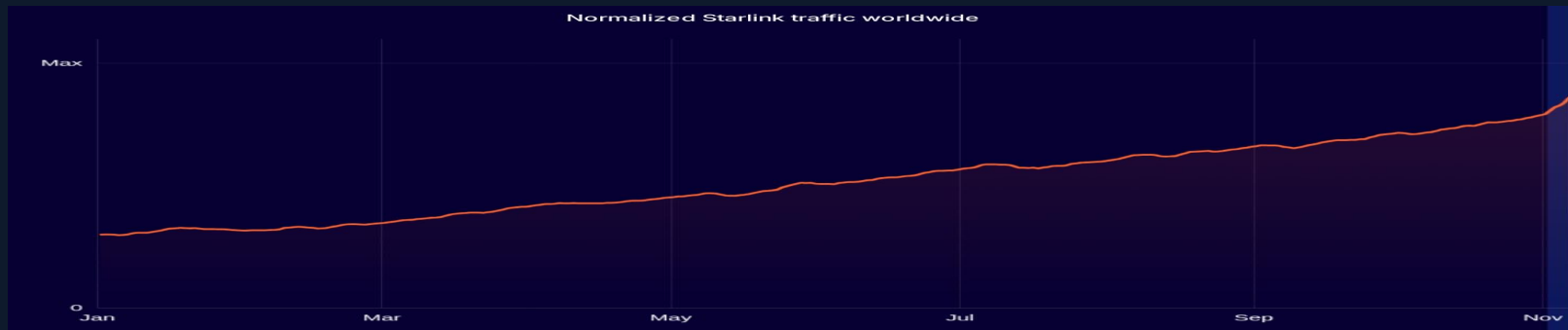
 **17%**

Global Internet traffic growth (YoY)

 **3.3x**

Starlink traffic growth (normalized)

Starlink normalized traffic trend (2024)



Note: Trend line starts mid-January to normalize for post-holiday activity; Starlink scaled relative to peak request volume.

Google Leads Top 10 Services as Generative AI Rises

Top 10 most popular Internet services (worldwide)

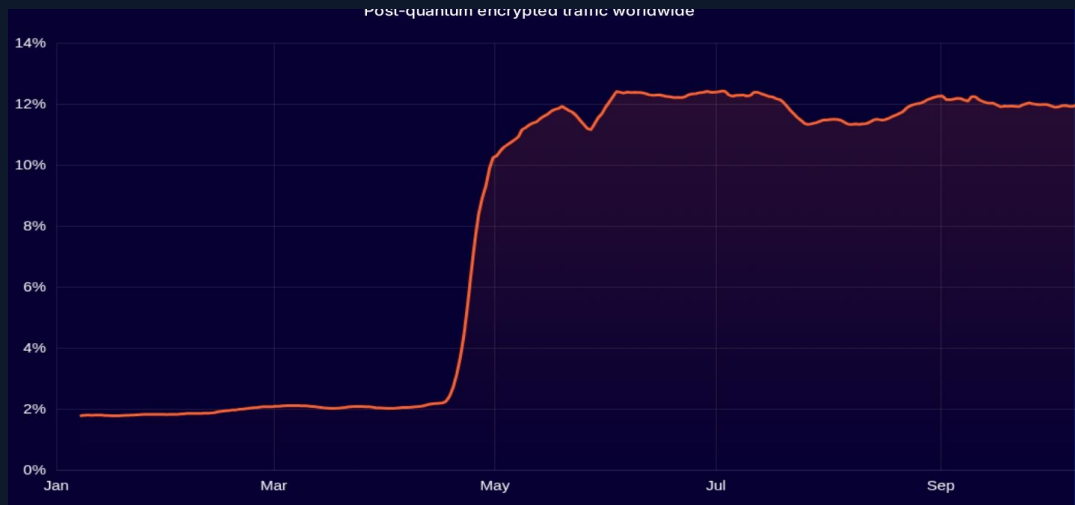
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Key notes

- Google ranked #1 overall (2nd consecutive year).
- Ranking based on aggregate data from Cloudflare 1.1.1.1 DNS resolver.
- Generative AI continued its rapid rise throughout 2024 (while the Metaverse faded).

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-quantum encrypted traffic (TLS 1.3)



~2% → ~12–13% following Chrome 124 default enablement (Apr 17, 2024).

Adoption highlights



HTTP/3
21% of traffic



API clients
Go leads (~12%)

Next: Node.js, Python (9.6%), Java (7.4%), .NET (3.6%)

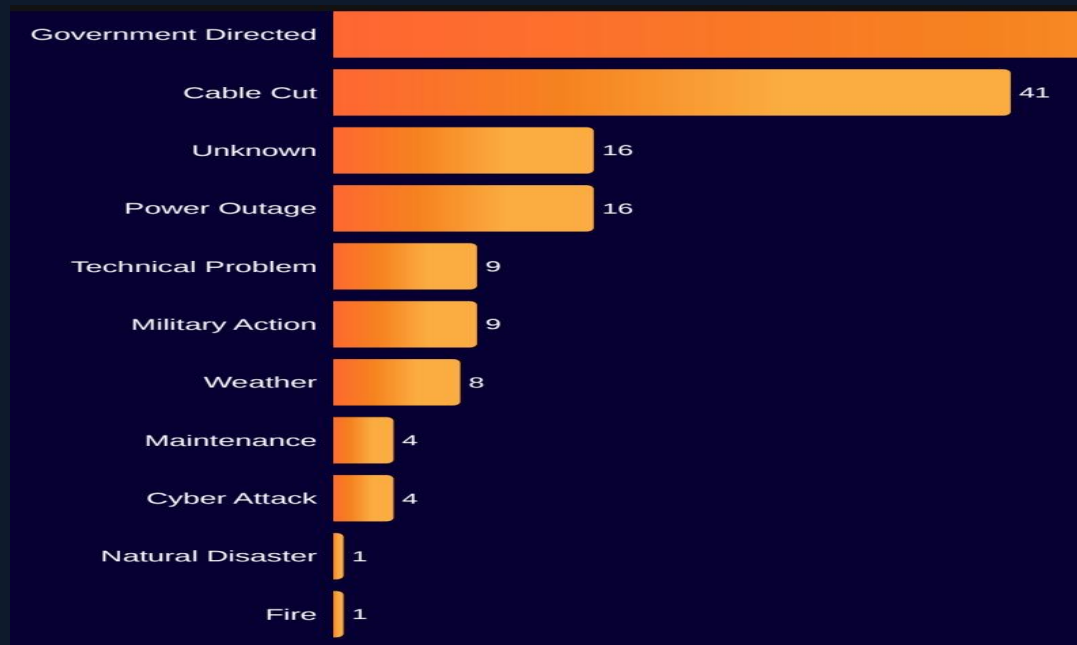


PQE momentum

Firefox rolling out; Safari testing

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Observed major Internet disruptions (2024): 225



Key drivers

- Government-directed shutdowns: 116+
- Cable cuts: 41
- Power outages: 16
- Unknown: 16
- Technical problems: 9;
Military action: 9

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

 **6.5%**

Traffic mitigated by Cloudflare

Most attacked sectors

Gambling and Games

Also: bot concentration (top 10 countries account for 68% of bot traffic)

Persistence: Log4j

CVE-2021-44228 continues to be actively targeted more than two years after disclosure (Dec 2021).

Routing security

RPKI valid routes +6.4%

Signal toward safer routing posture.

Email threats

4.3% malicious emails

43% of malicious emails contain a deceptive link.

Top malicious/spam TLD: .bar

Five Takeaways Shaping the 2025 Internet Outlook



Strategic takeaways

- Plan for sustained demand: 17% YoY traffic growth is now the baseline.
- Prepare crypto agility: post-quantum TLS 1.3 jumped from ~2% to ~13% after Chrome 124 (Apr 17, 2024).
- Harden for geopolitical fragility: government-directed shutdowns drove 116+ of 225 major disruptions.
- Operate assuming constant attack: Cloudflare mitigated 6.5% of traffic; Log4j exploitation persists.
- Align developer strategy to reality: Go leads API clients (~12%); HTTP/3 adoption reached 21%.